Solutions and teaching notes, chapter 6

Teacher use. The following answers concern the tasks not solved in the lesson. Do first make the approach explicit; provide graduated help before the solution. The durations are estimates to be calibrated in class.

Diagnostic and remediation

1. 6. *Action:* Go back to rate 2 $a+h$.
2. $\mathbb{R}\setminus\{2\}$. *Action:* Set the denominator to zero.
3. Positive outside [-2; 1], negative between the roots, zero at the endpoints. *Action:* Construct two sign rows.
4. x^3-2x^2+x-2. *Action:* Distribute each term without leaving any out.
5. They are equal. *Action:* Substitute -x in brackets.
6. $y=2$. *Action:* Go back to the tangent equation and simplify.

Investigation

x^3 is strictly increasing with a horizontal tangent at 0. For a differentiable increasing function, the rates on both sides are positive or zero, so the derivative cannot be negative.

Direct applications

1. $5x^4$ on $\mathbb{R}$; $-\dfrac{2}{x^3}$ for $x\neq 0$; $\dfrac{1}{2\sqrt{x}}$ for $x>0$.
2. $2x(3x-2)+3(x^2+1)=9x^2-4x+3$; $-\dfrac{2x}{(x^2+1)^2}$ on $\mathbb{R}$.
3. $f\prime=2x-4$, so decreasing up to 2, then increasing. Minimum -3 at 2 ; values at the endpoints 1 and -2, so the maximum is 1 at 0.
4. First even, second odd; difference $x^2+1-2x=(x-1)^2\geq 0$, equality at 1.
5. $A(x)=x(6-x)$ on [0; 6] ; $A\prime=6-2x$. Maximum 9 m ^{2} at 3.

Exercises: methods and answers

Exercise 6.1 · A collection of derivatives. $12x^3-2$; $2x(x-1)+(x^2+2)=3x^2-2x+2$; $\dfrac{[2(x+3)-(2x+1)]}{(x+3)^2}=\dfrac{5}{(x+3)^2}$ for $x\neq -3$.

Exercise 6.2 · A complete table. $f\prime=3(x-1)(x-3)$; signs + , −, +. Values $f(0)=1$, $f(1)=5$, $f(3)=1$, $f(4)=5$. Minimum 1 at 0 and 3; maximum 5 at 1 and 4. The endpoints count.

Exercise 6.3 · A quotient. $f\prime=1-\dfrac{4}{x^2}=\dfrac{(x-2)(x+2)}{x^2}$. Negative sign before 2, positive after. Minimum 4 at 2; for $x>0$, $x+\dfrac{4}{x}\geq 4$, equality only at 2.

Exercise 6.4 · Compare two curves. $f-g=x(x-1)(x+1)$. Signs: negative on $]-\infty;-1[$, positive on $]-1;0[$, negative on $]0;1[$, positive on $]1;+\infty[$. Intersection at $-1, 0, 1$. We can do the table of the three factors without differentiation.

Exercise 6.5 · A package. $V=x(12-2x)^2=4x(6-x)^2$. $V'=12(x-2)(x-6)$, positive on $[0;2[$, negative on $]2;6[$. $V(0)=V(6)=0$, $V(2)$ = 128 cm ^{3}. Maximum at x=2 cm; box 8 $\times 8 \times 2$.

Exercise 6.6 · Stationary without extremum. $f'=3x^2$, so the value at 0 is 0. In every neighbourhood of 0, negative x values give $f(x)<0$ and positive x $f(x)>0$; no extremum.

Exercise 6.7 · Compare strategies. Square: $(x-3)^2+3\geq 3$. Derivative: $2x-6$, minimum at 3, value 3. The first avoids a complete study of variations; the second illustrates the link between chapters.

Exercise 6.8 · Delayed review. It decreases on $]-\infty;0[$ and $]0;+\infty[$. It does not decrease on their union: $-1<1$ but $f(-1)=-1<1=f(1)$. The domain is not an interval.

CQFD, Exercise 12 p. 156, Questions 1 and 2. $f(x)-g(x)=\frac{1}{2}x^2-1=\frac{1}{2}(x-\sqrt{2})(x+\sqrt{2})$. The curve of f lies strictly above that of g for $x<-\sqrt{2}$ or $x>\sqrt{2}$, below between these two numbers; equality at the x-values $\pm\sqrt{2}$.

CQFD, exercise 16 p. 156, questions f and g. For $f(x)=x^3$, we can choose $]-\infty;0[$: $f(x)<0$ and the function is strictly increasing. For $g(x)=x^2-4$, we can choose $[0;2[$: $g(x)<0$ and the function is strictly increasing on this interval. Subintervals not reduced to a single point are also suitable.

MCQ: answers and reasons

MCQ 1.

1. B. Integer power.

2. A. Rate of change of the reciprocal.

3. D. Product rule.

4. C. Sign of the derivative and variation.

5. B. Non-zero denominator.

MCQ 2.

1. C. $f(-x)=-f(x)$.

2. A. The other conclusions need more.

3. D. Formula and domain.

4. B. Attained at 1.

5. C. Derivative $2x_0=0$.

True or false

1. False: $f(x) = \dfrac{1}{x} > 0$ on $]0; +\infty[$, but $f\prime(x) < 0$.

2. True, with the stated assumptions; at the endpoints, the conclusion is not forced.

3. False without sign assumptions: on $[-2; -1]$, $u(x) = v(x) = x$ increase but x^2 decreases.

Differentiated teaching. If a pupil is stuck on a technical point, suggest only the first step (domain, factor or reference sample space), then remove the help. For an independent student, ask for a second method and the handling of a boundary case. Orally, allow an initial rephrasing in the language mastered, then require a precise mathematical sentence in French.